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MTH401 ASSIGNMENT 1 SOLUTION SPRING 2022

Provide by VU Answer

Due Date: 6 June 2022

Total Marks: 10 Marks

Important Instructions:

Write your ID on the top of your solution file.

All students are directed to use the font and style of text as is used in this document.

Use Math Type or Equation Editor etc. for mathematical symbols and equations.

Submit assignment in MS Word Format Only.

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Question 1:

Solve the following differential equation by using the method of separation of variables.

$$\frac{dy}{dx} = (1 + e^{-x})(y^2 - 1).$$

Solution:

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$$\frac{dy}{dx} = (1+e^{-x})(y^2-1)$$

$$\frac{dy}{(y^2-1)} = (1+e^{-x})dx$$

$$\therefore \frac{1}{y^2-1} = \frac{A}{y+1} + \frac{B}{y-1} \rightarrow (eq=1)$$

Both side mutiply (y^2-1)

$$1 = A(y-1) + B(y+1)$$

put $y = -1$

$$1 = A(-1-1) + B(-1+1)$$

$$1 = A(-2) + B(0)$$

$$1 = -2A$$

$$A = \frac{1}{-2}$$

put $y = 1$

$$1 = A(1-1) + B(1+1)$$

$$1 = A(0) + B(2)$$

$$B = \frac{1}{2}$$

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put value A and B in eq=1

$$\frac{1}{(y^2-1)}=\frac{-1}{2(y+1)}+\frac{1}{2(y-1)}$$

Taking integration on both side

$$\int\frac{dy}{(y^2-1)}=\int(1+e^{-x})dx$$

$$\int\frac{-1}{2(y+1)}dy+\int\frac{1}{2(y-1)}dy=\int 1dx+\int e^{-x}dx$$

$$-\frac{1}{2}\ln|y+1|+\frac{1}{2}\ln|y-1|=x+\frac{e^{-x}}{-1}+c$$

$$-\ln(y+1)^{\frac{1}{2}}+\ln(y-1)^{\frac{1}{2}}=x-e^{-x}+c$$

$$\ln\frac{(y-1)^{\frac{1}{2}}}{(y+1)^{\frac{1}{2}}}=x-e^{-x}+c$$

taking anti log atham on both side

$$\frac{(y-1)^{\frac{1}{2}}}{(y+1)^{\frac{1}{2}}}=e^{x-e^{-x}}+c$$

$$\sqrt{\frac{(y-1)}{(y+1)}}=e^{x-e^{-x}}$$

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Question 2:

Check whether the following differential equation is exact or not. If not, then convert it into an exact differential equation.

$$2ydx + (x - sin\, y^{\frac{1}{2}})dy = 0.$$

Solution:

Check differential equation is exact or not

$\partial M=2y$

$$\frac{\partial M}{\partial y} = 2$$

$$\partial N = x - \sin y^{\frac{1}{2}}$$

$$\frac{\partial N}{\partial x} = 1$$

$$\frac{\partial M}{\partial N} \neq \frac{\partial N}{\partial x}$$

Not Exact

Then Now

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$$\exp\left[\int \frac{\frac{\partial N}{\partial y}-\frac{\partial M}{\partial x}}{M}dy\right]$$

$$\exp\left[\int \frac{1-2}{2y}dy\right]$$

$$= e^{\frac{1}{2}\int \frac{1}{y}dy}$$

$$= e^{\frac{-1}{2}\ln y}$$

$$= y^{\frac{-1}{2}}$$

Mutiply $y^{\frac{-1}{2}}$ in this factor

$$2y^{\frac{-1}{2}}dx + y^{\frac{-1}{2}}(x-\sin y^{\frac{1}{2}})dy = 0$$

$$2y^{\frac{1}{2}}dx + y^{\frac{-1}{2}}(x-\sin y^{\frac{1}{2}})dy = 0$$

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$$\partial M = 2y^{\frac{1}{2}}$$

$$\frac{\partial M}{\partial y} = 2\frac{1}{2}y^{\frac{-1}{2}}$$

$$\frac{\partial M}{\partial y} = y^{\frac{-1}{2}}$$

$$\partial N = y^{\frac{-1}{2}}x - y^{\frac{-1}{2}}\sin y^{\frac{1}{2}}$$

$$\frac{\partial N}{\partial x} = y^{\frac{-1}{2}}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Its become exact.

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If you found any mistake then correct yourself and inform me.

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